

MATH BOOKLET

June 9, 2026

1 Notes to Self

- ADAPTIVE RISE REMARK.
- Bounded weights REMARK
- REMARK about EPSILON PRIME
- REMARK ABOUT EXPONENTIAL CONVERGENCE

2 Lemmas

2.1 Notation/Lemmas

DEFINITION (Class $\mathcal{K}$, $\mathcal{KL}$, $\mathfrak{L}$, and $\mathcal{KL}_\infty$ functions) [?, Definition 3.3].

For $a_2 \in \overline{\mathbb{R}}_{>0}$, a function $\alpha : [0, a_2) \rightarrow \mathbb{R}_{\geq 0}$ belongs to class $\mathcal{K}$ (denoted $\alpha \in \mathcal{K}$) if it is (i) strictly increasing, (ii) continuous, and (iii) satisfies $\alpha(0) = 0$. A function $\alpha \in \mathcal{K}$ is also of class $\mathcal{K}_\infty$ (denoted $\alpha \in \mathcal{K}_\infty$) if $a_2 = \infty$ and $\lim_{s \rightarrow \infty} \alpha(s) = \infty$. A function $\ell : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ belongs to class $\mathfrak{L}$ (denoted $\ell \in \mathfrak{L}$) if it is (i) continuous, (ii) strictly decreasing, and (iii) $\lim_{s \rightarrow \infty} \ell(s) = 0$. A function $\beta : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is said to be of class $\mathcal{KL}$ (denoted $\beta \in \mathcal{KL}$) if it is (i) continuous, (ii) for each fixed $s \in \mathbb{R}_{\geq 0}$, the map $\mathbf{r} \mapsto \beta(\mathbf{r}, s)$ belongs to class $\mathcal{K}$, and (iii) for each fixed $\mathbf{r} \in \mathbb{R}_{\geq 0}$, $s \mapsto \beta(\mathbf{r}, s)$ belongs to class $\mathfrak{L}$. A function $\beta \in \mathcal{KL}$ also belongs to class $\mathcal{KL}_\infty$ (denoted $\beta \in \mathcal{KL}_\infty$) if for each fixed $s \in \mathbb{R}_{\geq 0}$, the map $\mathbf{r} \mapsto \beta(\mathbf{r}, s)$ is class $\mathcal{K}_\infty$ [?, Definition A.3].

Many of the functions in this work are similar to those of class $\mathcal{K}$ and class $\mathcal{K}_\infty$ but do not necessarily possess the property that $\alpha(0) = 0$. The following convention is introduced to avoid subsequent repetition.

DEFINITION (Class $\mathcal{P}$ functions).

Let $a_3 \in \overline{\mathbb{R}}_{>0}$. A continuous function $\rho : [0, a_3) \rightarrow \mathbb{R}_{\geq 0}$ is said to be of class $\mathcal{P}$ if it is strictly increasing. A function $\rho \in \mathcal{P}$ is also of class $\mathcal{P}_\infty$ (denoted $\rho \in \mathcal{P}_\infty$) if $a_3 = \infty$ and $\lim_{s \rightarrow \infty} (\rho(s)) = \infty$.

LEMMA

Let $D \subseteq \mathbb{R}^n$ be a convex, open set with $\mathbf{0}_n \in D$. Define $a_6 \triangleq \inf \{\|x\| \mid x \in \mathbb{R}^n \setminus D\}$ and let $\mathbf{r} \in [0, a_6)$ be such that $\overline{B}_{\mathbf{r}}(\mathbf{0}_n) \subsetneq D$.¹ Let $f_6 : D \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^m$ be locally Lipschitz continuous on D , uniformly in the second argument. Then, there exists a non-decreasing function $\sigma_1 : [0, a_6 - \mathbf{r}) \rightarrow \mathbb{R}_{\geq 0}$ such that for all $(x, x_d, t) \in D \times \overline{B}_{\mathbf{r}}(\mathbf{0}_n) \times \mathbb{R}_{\geq 0}$ that satisfy $\|x - x_d\| < a_6 - \mathbf{r}$,

$$\|f_6(x, t) - f_6(x_d, t)\| \leq \sigma_1(\|x - x_d\|) \|x - x_d\|. \quad (1)$$

LEMMA

For the map f_6 in Lemma 2.1, assume instead that $x \mapsto f_6(x, t)$ is of class $\mathcal{C}^1$ on D and that $h_0 : D \rightarrow \mathbb{R}_{\geq 0}$ defined as $h_0(x) \triangleq \sup_{t \in \mathbb{R}_{\geq 0}} \|\nabla_x f_6(x, t)\|$ is continuous on D . Then, (1) holds for all $(x, x_d, t) \in D \times \overline{B}_{\mathbf{r}}(\mathbf{0}_n) \times \mathbb{R}_{\geq 0}$ such that $\|x - x_d\| < a_6 - \mathbf{r}$ and $\sigma_1 : [0, a_6 - \mathbf{r}) \rightarrow \mathbb{R}_{\geq 0}$ is continuous.

REMARK

If f_6 is not a function of time for Lemma 2.1 (i.e., $x \mapsto f_6(x)$), the condition that h_0 must be continuous on D is not required, as this fact is implied by the requirement that f_6 be of class $\mathcal{C}^1$ on D .

¹If D is selected as an open ball, a_6 is its radius, i.e., $D = B_{a_6}(\mathbf{0}_n)$. If $D = \mathbb{R}^n$, let $a_6 = \infty$.

LEMMA

Let $a_8 \in \overline{\mathbb{R}}_{>0}$ and let $\sigma_2 : [0, a_8) \rightarrow \mathbb{R}_{\geq 0}$ be a continuous, non-decreasing function. Let $\rho_2(s) \triangleq \sigma_2(s) + cs$ for all $s \in [0, a_8)$ where $c \in \mathbb{R}_{>0}$ is a user-defined constant. Consequently, $\sigma_2(s) \leq \rho_2(s)$ for all $s \in [0, a_8)$ and $\rho_2 \in \mathcal{P}$. If $a_8 = \infty$, $\rho_2 \in \mathcal{P}_\infty$.

3 PROBLEM STATEMENT

Let the target agent's dynamics be

$$\ddot{q}_d = h(q_d, \dot{q}_d)$$

where $h \in \mathcal{C}^2$ is unknown (not necessarily linear-in-the-parameters). Let the controlled agent's dynamics be

$$\ddot{q} = f(q, \dot{q}) + u + d(t)$$

where $f \in \mathcal{C}^2$ is unknown (not necessarily linear-in-the-parameters). Assume q , $\dot{q}$, q_d , and $\dot{q}_d$ are known, $d \in \mathcal{C}^2$ is unknown, and $q_d \in \mathcal{C}^3$, where

$$\begin{aligned} \|d(t)\| &\leq \bar{d} \\ \|\dot{d}(t)\| &\leq \bar{\dot{d}} \\ \|\ddot{d}(t)\| &\leq \bar{\ddot{d}} \end{aligned}$$

are known constants. Assume

$$\begin{aligned} \|q_d(t)\| &\leq \bar{q}_d \\ \|\dot{q}_d(t)\| &\leq \bar{\dot{q}}_d \\ \|\ddot{q}_d(t)\| &\leq \bar{\ddot{q}}_d \\ \|\ddot{\ddot{q}}_d(t)\| &\leq \bar{\ddot{\ddot{q}}}_d \end{aligned}$$

are known constants. Define

$$\begin{aligned} e &\triangleq q_d - q \\ r_1 &\triangleq \dot{e} + k_1 e \\ r_2 &\triangleq \dot{r}_1 + k_2 r_1 \end{aligned}$$

Assume

$$\begin{aligned} \|\hat{\theta}\| &\leq \bar{\theta} \\ \|\theta^*\| &\leq \bar{\theta} \\ \|\dot{\hat{\theta}}\| &\leq \eta_1 \end{aligned}$$

are known constants. Let

$$z \triangleq \begin{bmatrix} e^\top & r_1^\top & r_2^\top \end{bmatrix}^\top \in \mathbb{R}^{3n}$$

Goal

Find a continuous u such that

$$z \rightarrow 0$$

exponentially.

4 Error Dynamics

E Error derivative

$$\dot{e} = r_1 - k_1 e$$

R1 Error derivative

$$\dot{r}_1 = r_2 - k_2 r_1$$

R2 Error derivative

1. First:

$$\begin{aligned}
\dot{r}_1 &= \ddot{e} + k_1 \dot{e} \\
&= \ddot{q}_d - \ddot{q} + k_1 (r_1 - k_1 e) \\
&= h(q_d, \dot{q}_d) - (f(q, \dot{q}) + u + d(t)) + k_1 r_1 - k_1^2 e \\
&= h(q_d, \dot{q}_d) - f(q, \dot{q}) - u - d(t) + k_1 r_1 - k_1^2 e
\end{aligned}$$

2. Second:

$$\begin{aligned}
\dot{r}_2 &= \ddot{r}_1 + k_2 \dot{r}_1 \\
&= \frac{d}{dt} (h(q_d, \dot{q}_d) - f(q, \dot{q}) - u - d(t) + k_1 r_1 - k_1^2 e) + k_2 \dot{r}_1 \\
&= \frac{\partial h}{\partial q_d} \Big|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial h}{\partial \dot{q}_d} \Big|_{(q_d, \dot{q}_d)} \ddot{q}_d - \frac{\partial f}{\partial q} \Big|_{(q, \dot{q})} \dot{q} - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} \ddot{q} - \dot{u} - \dot{d}(t) + k_1 \dot{r}_1 - k_1^2 \dot{e} + k_2 \dot{r}_1 \\
&= \frac{\partial h}{\partial q_d} \Big|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial h}{\partial \dot{q}_d} \Big|_{(q_d, \dot{q}_d)} h(q_d, \dot{q}_d) - \frac{\partial f}{\partial q} \Big|_{(q, \dot{q})} \dot{q} - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} (f(q, \dot{q}) + u + d(t)) \\
&\quad - \dot{u} - \dot{d}(t) + k_1 \dot{r}_1 - k_1^2 (r_1 - k_1 e) + k_2 \dot{r}_1 \\
&= \underbrace{\frac{\partial h}{\partial q_d} \Big|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial h}{\partial \dot{q}_d} \Big|_{(q_d, \dot{q}_d)} h(q_d, \dot{q}_d) - \frac{\partial f}{\partial q} \Big|_{(q, \dot{q})} \dot{q} - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} f(q, \dot{q}) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} u}_{F(\kappa)} \\
&\quad - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{u} - \dot{d}(t) + (k_1 + k_2) \dot{r}_1 - k_1^2 r_1 + k_1^3 e \\
&= F(\kappa) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{u} - \dot{d}(t) + (k_1 + k_2) (r_2 - k_2 r_1) - k_1^2 r_1 + k_1^3 e \\
&= F(\kappa) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{u} - \dot{d}(t) + (k_1 + k_2) r_2 - (k_1 + k_2) k_2 r_1 - k_1^2 r_1 + k_1^3 e \\
&= F(\kappa) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{u} - \dot{d}(t) + (k_1 + k_2) r_2 + (-k_1 k_2 - k_2^2 - k_1^2) r_1 + k_1^3 e
\end{aligned}$$

3. Assume that the activation functions are chosen such that $\Phi \in \mathcal{C}^2(\mathbb{R}^{5n} \times \mathbb{R}^p; \mathbb{R}^n)$. NN (ResNet here) Approximation (where F and ε are $\mathcal{C}^1$ since h and f are $\mathcal{C}^2$). NOTE: u_d is the “ideal” controller that you would choose if you were already on the desired trajectory

$$\begin{aligned}
\varepsilon(\kappa) &\triangleq F(\kappa) - \Phi(\kappa, \theta^*) \\
\kappa &\triangleq \begin{bmatrix} q^\top & \dot{q}^\top & q_d^\top & \dot{q}_d^\top & u^\top \end{bmatrix}^\top \in \mathbb{R}^{5n} \\
\kappa_d &\triangleq \begin{bmatrix} q_d^\top & \dot{q}_d^\top & q_d^\top & \dot{q}_d^\top & u_d^\top \end{bmatrix}^\top \in \mathbb{R}^{5n} \\
u_d &\triangleq d(t) + f(q_d, \dot{q}_d) - \ddot{q}_d
\end{aligned}$$

4. Controller

$$\dot{u} = (k_1 + k_2 + k_3) r_2 + (1 - k_1^2 - k_2^2 - k_1 k_2) r_1 + k_1^3 e + K_{\text{RISE}} \text{sgn}(r_1) + \Phi(\kappa, \hat{\theta})$$

5. Third:

$$\begin{aligned}
\dot{r}_2 &= \Phi(\kappa, \theta^*) + \varepsilon(\kappa) - \left((k_1 + k_2 + k_3) r_2 + (1 - k_1^2 - k_2^2 - k_1 k_2) r_1 + k_1^3 e + K_{\text{RISE}} \text{sgn}(r_1) + \Phi(\kappa, \hat{\theta}) \right) \\
&\quad - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) + (k_1 + k_2) r_2 + (-k_1 k_2 - k_2^2 - k_1^2) r_1 + k_1^3 e \\
&= \Phi(\kappa, \theta^*) + \varepsilon(\kappa) - \left((\cancel{k_1 + k_2} + k_3) r_2 + \left(1 - \cancel{k_1^2 - k_2^2 - k_1 k_2} \right) r_1 + \cancel{k_1^3 e} + K_{\text{RISE}} \text{sgn}(r_1) + \Phi(\kappa, \hat{\theta}) \right) \\
&\quad - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) + \cancel{(k_1 + k_2) r_2} + \cancel{(-k_1 k_2 - k_2^2 - k_1^2) r_1} + \cancel{k_1^3 e} \\
&= \Phi(\kappa, \theta^*) + \varepsilon(\kappa) - \left(k_3 r_2 + r_1 + K_{\text{RISE}} \text{sgn}(r_1) + \Phi(\kappa, \hat{\theta}) \right) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) \\
&= \Phi(\kappa, \theta^*) + \varepsilon(\kappa) - k_3 r_2 - r_1 - K_{\text{RISE}} \text{sgn}(r_1) - \Phi(\kappa, \hat{\theta}) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) \\
&= \underbrace{\Phi(\kappa, \theta^*) - \Phi(\kappa, \hat{\theta})}_{N_0(\kappa, \theta^*, \hat{\theta})} + \varepsilon(\kappa) - k_3 r_2 - r_1 - K_{\text{RISE}} \text{sgn}(r_1) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) \\
&\quad - \varepsilon(\kappa_d) + \varepsilon(\kappa_d) + N_0(\kappa_d, \theta^*, \hat{\theta}) - N_0(\kappa_d, \theta^*, \hat{\theta}) + \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) \\
&= \underbrace{N_0(\kappa, \theta^*, \hat{\theta}) - N_0(\kappa_d, \theta^*, \hat{\theta})}_{\tilde{N}_1(\kappa, \kappa_d, \theta^*, \hat{\theta})} + \underbrace{\varepsilon(\kappa) - \varepsilon(\kappa_d)}_{\tilde{N}_\varepsilon(\kappa, \kappa_d)} - k_3 r_2 - r_1 - K_{\text{RISE}} \text{sgn}(r_1) \\
&\quad + \underbrace{\left(\frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} \right)}_{\tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d)} d(t) - \dot{d}(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) + \varepsilon(\kappa_d) + N_0(\kappa_d, \theta^*, \hat{\theta}) \\
&= \underbrace{\tilde{N}_1(\kappa, \kappa_d, \theta^*, \hat{\theta}) + \tilde{N}_\varepsilon(\kappa, \kappa_d) + \tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d) d(t)}_{\tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t)} - k_3 r_2 - r_1 - K_{\text{RISE}} \text{sgn}(r_1) \\
&\quad - \underbrace{\dot{d}(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) + \varepsilon(\kappa_d) + N_0(\kappa_d, \theta^*, \hat{\theta})}_{D(\kappa_d, \theta^*, \hat{\theta}, t)} \\
&= \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - k_3 r_2 - r_1 - K_{\text{RISE}} \text{sgn}(r_1) + D(\kappa_d, \theta^*, \hat{\theta}, t)
\end{aligned}$$

6. Let $K[\cdot]$ denote the Krasovskii regularization (or Filippov—doesn't matter here). Regularizing the above so we can study Filippov solutions to the resulting ensemble differential inclusion:

$$\dot{r}_2 \in \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - k_3 r_2 - r_1 - K_{\text{RISE}} K[\text{sgn}](r_1) + D(\kappa_d, \theta^*, \hat{\theta}, t)$$

5 Implementable Controller

Recall

$$\begin{aligned} e &\triangleq q_d - q \\ r_1 &\triangleq \dot{e} + k_1 e \\ \dot{r}_1 &= \ddot{e} + k_1 \dot{e} \\ r_2 &\triangleq \dot{r}_1 + k_2 r_1 \end{aligned}$$

Simplify:

$$\begin{aligned} \dot{u} &= (k_1 + k_2 + k_3) r_2 + (1 - k_1^2 - k_2^2 - k_1 k_2) r_1 + k_1^3 e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \\ &= (k_1 + k_2 + k_3) (\dot{r}_1 + k_2 r_1) + (1 - k_1^2 - k_2^2 - k_1 k_2) (\dot{e} + k_1 e) + k_1^3 e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \\ &= (k_1 + k_2 + k_3) (\ddot{e} + k_1 \dot{e} + k_2 (\dot{e} + k_1 e)) + (1 - k_1^2 - k_2^2 - k_1 k_2) (\dot{e} + k_1 e) + k_1^3 e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \\ &= (k_1 + k_2 + k_3) (\ddot{e} + k_1 \dot{e} + k_2 \dot{e} + k_1 k_2 e) + (1 - k_1^2 - k_2^2 - k_1 k_2) \dot{e} + (1 - k_1^2 - k_2^2 - k_1 k_2) k_1 e + k_1^3 e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \\ &= ((k_1 + k_2 + k_3) \ddot{e} + (k_1 + k_2 + k_3) k_1 \dot{e} + (k_1 + k_2 + k_3) k_2 \dot{e} + (k_1 + k_2 + k_3) k_1 k_2 e) \\ &\quad + (1 - k_1^2 - k_2^2 - k_1 k_2) \dot{e} + (k_1 - k_1^3 - k_1 k_2^2 - k_1^2 k_2) e + k_1^3 e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \\ &= (k_1 + k_2 + k_3) \ddot{e} + (k_1^2 + k_1 k_2 + k_1 k_3) \dot{e} + (k_1 k_2 + k_2^2 + k_2 k_3) \dot{e} + (k_1^2 k_2 + k_1 k_2^2 + k_1 k_2 k_3) e \\ &\quad + (1 - k_1^2 - k_2^2 - k_1 k_2) \dot{e} + (k_1 - k_1^3 - k_1 k_2^2 - k_1^2 k_2) e + k_1^3 e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \\ &= (k_1 + k_2 + k_3) \ddot{e} + (k_1^2 + k_1 k_2 + k_1 k_3 + 1 - k_1^2 - k_2^2 - k_1 k_2 + k_1 k_2 + k_2^2 + k_2 k_3) \dot{e} \\ &\quad + (k_1^2 k_2 + k_1 k_2^2 + k_1 k_2 k_3 + k_1 - k_1^3 - k_1 k_2^2 - k_1^2 k_2 + k_1^3) e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \\ &= (k_1 + k_2 + k_3) \ddot{e} + \left(\cancel{k_1^2} + k_1 k_2 + k_1 k_3 + 1 - \cancel{k_1^2} - \cancel{k_2^2 - k_1 k_2 + k_1 k_2} + \cancel{k_2^2} + k_2 k_3 \right) \dot{e} \\ &\quad + \left(\cancel{k_1^2 k_2} + \cancel{k_1 k_2^2} + k_1 k_2 k_3 + k_1 - \cancel{k_1^3} - \cancel{k_1 k_2^2} - \cancel{k_1^2 k_2} + \cancel{k_1^3} \right) e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \\ &= (k_1 + k_2 + k_3) \ddot{e} + (k_1 k_2 + k_1 k_3 + k_2 k_3 + 1) \dot{e} + (k_1 k_2 k_3 + k_1) e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \end{aligned}$$

Integrate

$$\begin{aligned} \int_{t_0}^t \dot{u}(\tau) d\tau &= \int_{t_0}^t \left((k_1 k_2 k_3 + k_1) e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \right) d\tau \\ &\quad + \int_{t_0}^t ((k_1 + k_2 + k_3) \ddot{e} + (k_1 k_2 + k_1 k_3 + k_2 k_3 + 1) \dot{e}) d\tau \\ u(t) - u(t_0) &= \int_{t_0}^t \left((k_1 k_2 k_3 + k_1) e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \right) d\tau \\ &\quad + (k_1 + k_2 + k_3) (\dot{e}(t) - \dot{e}(t_0)) + (k_1 k_2 + k_1 k_3 + k_2 k_3 + 1) (e(t) - e(t_0)) \end{aligned}$$

Let $u(t_0)$ cancel the other initial conditions: $u(t_0) = (k_1 + k_2 + k_3) \dot{e}(t_0) + (k_1 k_2 + k_1 k_3 + k_2 k_3 + 1) e(t_0)$. So,

$$u(t) = (k_1 + k_2 + k_3) \dot{e}(t) + (k_1 k_2 + k_1 k_3 + k_2 k_3 + 1) e(t) + \int_{t_0}^t \left(K_{\text{RISE}} \text{sgn}(r_1(\tau)) + \Phi(\kappa(\tau), \hat{\theta}(\tau)) + (k_1 k_2 k_3 + k_1) e(\tau) \right) d\tau$$

Simplify:

$$u(t) = K_P e(t) + K_D \dot{e}(t) + \int_{t_0}^t \left(K_I e(\tau) + K_{\text{RISE}} \text{sgn}(r_1(\tau)) + \Phi(\kappa(\tau), \hat{\theta}(\tau)) \right) d\tau$$

where

$$K_P \triangleq k_1 k_2 + k_1 k_3 + k_2 k_3 + 1$$

$$K_I \triangleq k_1 k_2 k_3 + k_1$$

$$K_D \triangleq k_1 + k_2 + k_3$$

6 Bounding Lemmas

ASSUMPTION.

Wherever a bound function is introduced below, it is assumed that this function (or a more conservative bound) is known.

BOUNDING LEMMA 1.1

$$\begin{aligned} \left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| &\leq \|e\| + \|\dot{e}\| \\ &\leq \|e\| + \|r_1 - k_1 e\| \\ &\leq \|e\| + \|r_1\| + k_1 \|e\| \\ &\leq (k_1 + 1) \|z\| \end{aligned}$$

BOUNDING LEMMA 1.2

Since q_d and $\dot{q}_d$ are bounded and f is class $\mathcal{C}^1$ ($\mathcal{C}^2$ in fact), by Lemma 2.1, there exists a $\mathcal{C}^0$ ($\mathcal{C}^1$ in fact) nondecreasing function $\sigma_f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\begin{aligned} \|f(q, \dot{q}) - f(q_d, \dot{q}_d)\| &\leq \sigma_f \left(\left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| \right) \left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| \\ \sigma_f(s) &\triangleq \sup_{\begin{bmatrix} q^\top & \dot{q}^\top \end{bmatrix}^\top \in \bar{B}_{\bar{q}_d + \bar{\dot{q}}_d + s}(0_{2n})} \left(\left\| \frac{df}{d \begin{bmatrix} q^\top & \dot{q}^\top \end{bmatrix}^\top} \right|_{(q, \dot{q})} \right\| \end{aligned}$$

By Lemma 2.1, there exists a class $\mathcal{P}_\infty$ function $\rho_f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that (for some $c > 0$)

$$\begin{aligned} \|f(q, \dot{q}) - f(q_d, \dot{q}_d)\| &\leq \rho_{f1} \left(\left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| \right) \left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| \\ \rho_{f1}(s) &\triangleq \sigma_f(s) + cs \end{aligned}$$

By Bounding Lemma 1.1,

$$\begin{aligned} \|f(q, \dot{q}) - f(q_d, \dot{q}_d)\| &\leq \rho_{f1} \left(\left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| \right) \left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| \\ &\leq \underbrace{\rho_{f1}((k_1 + 1) \|z\|) (k_1 + 1) \|z\|}_{\rho_f(k_1, \|z\|)} \\ &\leq \rho_f(k_1, \|z\|) \|z\| \end{aligned}$$

BOUNDING LEMMA 1.3

Recall

$$\tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d) \triangleq \left. \frac{\partial f}{\partial \dot{q}} \right|_{(q_d, \dot{q}_d)} - \left. \frac{\partial f}{\partial \dot{q}} \right|_{(q, \dot{q})}$$

Since q_d and $\dot{q}_d$ are bounded and $\frac{\partial f}{\partial \dot{q}}$ is class $\mathcal{C}^1$, by Lemma 2.1, there exists a $\mathcal{C}^0$ nondecreasing function $\sigma_{f'} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\left\| \left. \frac{\partial f}{\partial \dot{q}} \right|_{(q_d, \dot{q}_d)} - \left. \frac{\partial f}{\partial \dot{q}} \right|_{(q, \dot{q})} \right\| \leq \sigma_{f'} \left(\left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| \right) \left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\|$$

$$\sigma_{f'}(s) \triangleq \sup_{\begin{bmatrix} q^\top & \dot{q}^\top \end{bmatrix}^\top \in \overline{B}_{\bar{q}_d + \bar{q}_d + s}(0_{2n})} \left(\left\| \left. \frac{\partial^2 f}{\partial \dot{q}^2} \right|_{(q, \dot{q})} \right\| + \left\| \left. \frac{\partial^2 f}{\partial \dot{q} \partial q} \right|_{(q, \dot{q})} \right\| \right)$$

By Lemma 2.1, there exists a class $\mathcal{P}_\infty$ function $\rho_{f'} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that (for some $c > 0$)

$$\left\| \tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d) \right\| \leq \rho_{f'} \left(\left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\| \right) \left\| \begin{bmatrix} e^\top & \dot{e}^\top \end{bmatrix}^\top \right\|$$

$$\rho_{f'}(s) \triangleq \sigma_{f'}(s) + cs$$

By Bounding Lemma 1.1,

$$\left\| \tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d) \right\| \leq \underbrace{\rho_{f'}((k_1 + 1) \|z\|) (k_1 + 1) \|z\|}_{\rho_{f'}(k_1, \|z\|)}$$

$$\leq \rho_{f'}(k_1, \|z\|) \|z\|$$

BOUNDING LEMMA 2.1

The ideal controller u_d satisfies

$$\ddot{q}_d = f(q_d, \dot{q}_d) + u_d + d(t)$$

$$\ddot{q}_d - f(q_d, \dot{q}_d) - d(t) = u_d$$

$$u_d = \ddot{q}_d - f(q_d, \dot{q}_d) - d(t)$$

By the stated assumptions and EVT, there exists a known constant

$$\bar{u}_d \triangleq \bar{\ddot{q}}_d + \sup_{(q_d, \dot{q}_d) \in \overline{B}_{\bar{q}_d}(0_n) \times \overline{B}_{\bar{q}_d}(0_n)} (\|f(q_d, \dot{q}_d)\|) + \bar{d}$$

such that

$$\|u_d\| \leq \bar{u}_d$$

BOUNDING LEMMA 2.2

Taking the derivative of the ideal controller gives

$$\dot{u}_d = \ddot{\ddot{q}}_d - \left. \frac{\partial f}{\partial q} \right|_{(q_d, \dot{q}_d)} \dot{q}_d - \left. \frac{\partial f}{\partial \dot{q}} \right|_{(q_d, \dot{q}_d)} \ddot{q}_d - \dot{d}(t)$$

By the stated assumptions and EVT, there exists a known constant

$$\bar{\dot{u}}_d \triangleq \bar{\ddot{\ddot{q}}}_d + \sup_{(q_d, \dot{q}_d) \in \overline{B}_{\bar{q}_d}(0_n) \times \overline{B}_{\bar{q}_d}(0_n)} \left(\left\| \left. \frac{\partial f}{\partial q} \right|_{(q_d, \dot{q}_d)} \right\| \right) \bar{\ddot{q}}_d + \sup_{(q_d, \dot{q}_d) \in \overline{B}_{\bar{q}_d}(0_n) \times \overline{B}_{\bar{q}_d}(0_n)} \left(\left\| \left. \frac{\partial f}{\partial \dot{q}} \right|_{(q_d, \dot{q}_d)} \right\| \right) \bar{\ddot{q}}_d + \bar{\dot{d}}$$

such that

$$\|\dot{u}_d\| \leq \bar{\dot{u}}_d$$

BOUNDING LEMMA 2.3

Recall

$$\kappa_d \triangleq \begin{bmatrix} q_d^\top & \dot{q}_d^\top & q_d^\top & \dot{q}_d^\top & u_d^\top \end{bmatrix}^\top \in \mathbb{R}^{5n}$$

Thus, by Bounding Lemma 2.1,

$$\|\kappa_d\| \leq \bar{\kappa}_d$$

$$\bar{\kappa}_d \triangleq 2\bar{q}_d + 2\bar{\dot{q}}_d + \bar{u}_d$$

BOUNDING LEMMA 2.4

Define

$$\tilde{u} \triangleq u_d - u$$

Recall u_d satisfies

$$\ddot{q}_d = d(t) + f(q_d, \dot{q}_d) + u_d$$

Recall u satisfies

$$\ddot{q} = d(t) + f(q, \dot{q}) + u$$

Subtract them

$$\begin{aligned}\ddot{q}_d - \ddot{q} &= f(q_d, \dot{q}_d) + u_d - f(q, \dot{q}) - u \\ \ddot{e} &= f(q_d, \dot{q}_d) + \tilde{u} - f(q, \dot{q}) \\ \ddot{e} - f(q_d, \dot{q}_d) + f(q, \dot{q}) &= \tilde{u} \\ \tilde{u} &= \ddot{e} - f(q_d, \dot{q}_d) + f(q, \dot{q})\end{aligned}$$

Recall

$$\begin{aligned}r_1 &\triangleq \dot{e} + k_1 e \\ \dot{r}_1 &= \ddot{e} + k_1 \dot{e} \\ r_2 &\triangleq \dot{r}_1 + k_2 r_1\end{aligned}$$

Thus,

$$\begin{aligned}r_2 &= \ddot{e} + k_1 \dot{e} + k_2 r_1 \\ r_2 - k_1 \dot{e} - k_2 r_1 &= \ddot{e} \\ \ddot{e} &= r_2 - k_1 \dot{e} - k_2 r_1\end{aligned}$$

Thus,

$$\tilde{u} = r_2 - k_1 \dot{e} - k_2 r_1 - f(q_d, \dot{q}_d) + f(q, \dot{q})$$

By Bounding Lemma 1.2,

$$\begin{aligned}\|\tilde{u}\| &\leq \|r_2\| + k_1 \|\dot{e}\| + k_2 \|r_1\| + \|f(q, \dot{q}) - f(q_d, \dot{q}_d)\| \\ &\leq \|r_2\| + k_1 \|r_1 - k_1 e\| + k_2 \|r_1\| + \rho_f(k_1, \|z\|) \|z\| \\ &\leq \|r_2\| + k_1 \|r_1\| + k_1^2 \|e\| + k_2 \|r_1\| + \rho_f(k_1, \|z\|) \|z\| \\ &\leq \|z\| + k_1 \|z\| + k_1^2 \|z\| + k_2 \|z\| + \rho_f(k_1, \|z\|) \|z\| \\ &\leq \underbrace{\left(1 + k_1 + k_1^2 + k_2 + \rho_f(k_1, \|z\|)\right)}_{\rho_{\tilde{u}}(k_1, k_2, \|z\|)} \|z\| \\ \|\tilde{u}\| &\leq \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\|\end{aligned}$$

BOUNDING LEMMA 2.5

Recall

$$\begin{aligned}\kappa &\triangleq \begin{bmatrix} q^\top & \dot{q}^\top & q_d^\top & \dot{q}_d^\top & u^\top \end{bmatrix}^\top \in \mathbb{R}^{5n} \\ \kappa_d &\triangleq \begin{bmatrix} q_d^\top & \dot{q}_d^\top & q_d^\top & \dot{q}_d^\top & u_d^\top \end{bmatrix}^\top \in \mathbb{R}^{5n} \\ \tilde{u} &\triangleq u_d - u\end{aligned}$$

Let

$$\begin{aligned}\tilde{\kappa} &\triangleq \kappa_d - \kappa \\ &= \begin{bmatrix} q_d^\top & \dot{q}_d^\top & q_d^\top & \dot{q}_d^\top & u_d^\top \end{bmatrix}^\top - \begin{bmatrix} q^\top & \dot{q}^\top & q_d^\top & \dot{q}_d^\top & u^\top \end{bmatrix}^\top \\ &= \begin{bmatrix} e^\top & \dot{e}^\top & 0_{1 \times n} & 0_{1 \times n} & \tilde{u}^\top \end{bmatrix}^\top\end{aligned}$$

By Bounding Lemma 2.4,

$$\begin{aligned}
\|\tilde{\kappa}\| &\leq \|e\| + \|\dot{e}\| + \|\tilde{u}\| \\
&\leq \|e\| + \|r_1 - k_1 e\| + \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\| \\
&\leq \|e\| + \|r_1\| + k_1 \|e\| + \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\| \\
&\leq (k_1 + 2) \|z\| + \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\| \\
&\leq \left(\underbrace{k_1 + 2 + \rho_{\tilde{u}}(k_1, k_2, \|z\|)}_{\rho_{\tilde{\kappa}}(k_1, k_2, \|z\|)} \right) \|z\| \\
&\leq \rho_{\tilde{\kappa}}(k_1, k_2, \|z\|) \|z\|
\end{aligned}$$

BOUNDING LEMMA 2.6

Notice

$$\dot{\kappa}_d = \begin{bmatrix} \dot{q}_d^\top & \ddot{q}_d^\top & \dot{q}_d^\top & \ddot{q}_d^\top & \dot{u}_d^\top \end{bmatrix}^\top \in \mathbb{R}^{5n}$$

By Bounding Lemma 2.2,

$$\begin{aligned}
\|\dot{\kappa}_d\| &\leq \bar{\kappa}_d \\
\bar{\kappa}_d &\triangleq 2\bar{q}_d + 2\bar{\ddot{q}}_d + \bar{u}_d
\end{aligned}$$

BOUNDING LEMMA 3.1

Recall

$$\tilde{N}_\varepsilon(\kappa, \kappa_d) \triangleq \varepsilon(\kappa) - \varepsilon(\kappa_d)$$

Since κ_d is bounded (by Bounding Lemma 2.3) and $\varepsilon \in \mathcal{C}^1$, by Lemma 2.1, there exists an **unknown** nondecreasing $\mathcal{C}^0$ function $\sigma_\varepsilon : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\begin{aligned}
\|\tilde{N}_\varepsilon(\kappa, \kappa_d)\| &\leq \sigma_{\varepsilon 1}(\|\tilde{\kappa}\|) \|\tilde{\kappa}\| \\
\sigma_{\varepsilon 1}(s) &\triangleq \sup_{\kappa \in \bar{B}_{\bar{\kappa}_d + s}(\mathbf{0}_n)} \left(\left\| \frac{d\varepsilon}{d\kappa} \Big|_\kappa \right\| \right)
\end{aligned}$$

Thus,

$$\begin{aligned}
\|\tilde{N}_\varepsilon(\kappa, \kappa_d)\| &\leq \underbrace{\sigma_{\varepsilon 1}(\rho_{\tilde{\kappa}}(k_1, k_2, \|z\|) \|z\|)}_{\sigma_\varepsilon(k_1, k_2, \|z\|)} \rho_{\tilde{\kappa}}(k_1, k_2, \|z\|) \|z\| \\
&\leq \sigma_\varepsilon(k_1, k_2, \|z\|) \|z\|
\end{aligned}$$

BOUNDING LEMMA 3.2

Recall

$$\|\theta\| \leq \bar{\theta}$$

Consider

$$\Phi(\kappa_d, \theta) - \Phi(\kappa, \theta)$$

Since Φ is $\mathcal{C}^1$ ($\mathcal{C}^2$ in fact) and κ_d is bounded, by Lemma 2.1, there exists a $\mathcal{C}^0$ ($\mathcal{C}^1$ in fact) nondecreasing function $\sigma_{\Phi\kappa} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\begin{aligned}
\|\Phi(\kappa_d, \theta) - \Phi(\kappa, \theta)\| &\leq \sigma_{\Phi\kappa}(\|\tilde{\kappa}\|) \|\tilde{\kappa}\| \\
\sigma_{\Phi\kappa}(s) &\triangleq \sup_{\|\theta\| \leq \bar{\theta}, \|\zeta\| \leq \bar{\kappa}_d + s} \left(\left\| \frac{\partial \Phi}{\partial \kappa} \Big|_{(\zeta, \theta)} \right\| \right)
\end{aligned}$$

By Lemma 2.1, there exists a class $\mathcal{P}_\infty$ function $\rho_{\Phi\kappa} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that (for some $c > 0$)

$$\begin{aligned}
\|\Phi(\kappa_d, \theta) - \Phi(\kappa, \theta)\| &\leq \rho_{\Phi\kappa}(\|\tilde{\kappa}\|) \|\tilde{\kappa}\| \\
\rho_{\Phi\kappa}(s) &\triangleq \sigma_{\Phi\kappa}(s) + cs
\end{aligned}$$

By Bounding Lemma 2.5,

$$\begin{aligned} \|\Phi(\kappa_d, \theta) - \Phi(\kappa, \theta)\| &\leq \underbrace{\rho_{\Phi\kappa}(\rho_{\tilde{\kappa}}(k_1, k_2, \|z\|) \|z\|) \rho_{\tilde{\kappa}}(k_1, k_2, \|z\|)}_{\rho_{\Phi\kappa}(k_1, k_2, \|z\|)} \|z\| \\ &\leq \rho_{\Phi\kappa}(k_1, k_2, \|z\|) \|z\| \end{aligned}$$

BOUNDING LEMMA 3.3

Recall

$$\begin{aligned} \tilde{N}_1(\kappa, \kappa_d, \theta^*, \hat{\theta}) &\triangleq N_0(\kappa, \theta^*, \hat{\theta}) - N_0(\kappa_d, \theta^*, \hat{\theta}) \\ N_0(\kappa, \theta^*, \hat{\theta}) &\triangleq \Phi(\kappa, \theta^*) - \Phi(\kappa, \hat{\theta}) \end{aligned}$$

So,

$$\begin{aligned} \tilde{N}_1(\kappa, \kappa_d, \theta^*, \hat{\theta}) &= \Phi(\kappa, \theta^*) - \Phi(\kappa, \hat{\theta}) - \Phi(\kappa_d, \theta^*) + \Phi(\kappa_d, \hat{\theta}) \\ &= \Phi(\kappa_d, \hat{\theta}) - \Phi(\kappa, \hat{\theta}) + \Phi(\kappa, \theta^*) - \Phi(\kappa_d, \theta^*) \end{aligned}$$

By triangle inequality and Bounding Lemma 3.2 (and since $\|\theta^*\| \leq \bar{\theta}$ and $\|\hat{\theta}\| \leq \bar{\theta}$), there exists class $\mathcal{P}_\infty$ functions $\rho_{\hat{\Phi}\kappa}, \rho_{\Phi^*\kappa} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\begin{aligned} \|\tilde{N}_1(\kappa, \kappa_d, \theta^*, \hat{\theta})\| &\leq \|\Phi(\kappa_d, \hat{\theta}) - \Phi(\kappa, \hat{\theta})\| + \|\Phi(\kappa, \theta^*) - \Phi(\kappa_d, \theta^*)\| \\ &\leq \rho_{\hat{\Phi}\kappa}(k_1, k_2, \|z\|) \|z\| + \rho_{\Phi^*\kappa}(k_1, k_2, \|z\|) \|z\| \\ &\leq \left(\underbrace{\rho_{\hat{\Phi}\kappa}(k_1, k_2, \|z\|) + \rho_{\Phi^*\kappa}(k_1, k_2, \|z\|)}_{\rho_{\Phi}(k_1, k_2, \|z\|)} \right) \|z\| \\ &\leq \rho_{\Phi}(k_1, k_2, \|z\|) \|z\| \end{aligned}$$

BOUNDING LEMMA 3.4

Recall

$$\tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) \triangleq \tilde{N}_1(\kappa, \kappa_d, \theta^*, \hat{\theta}) + \tilde{N}_\varepsilon(\kappa, \kappa_d) + \tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d) d(t)$$

By Bounding Lemma 1.3, 3.1, and 3.3,

$$\begin{aligned} \|\tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t)\| &\leq \|\tilde{N}_1(\kappa, \kappa_d, \theta^*, \hat{\theta})\| + \|\tilde{N}_\varepsilon(\kappa, \kappa_d)\| + \|\tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d)\| \|d(t)\| \\ &\leq \rho_{\Phi}(k_1, k_2, \|z\|) \|z\| + \sigma_\varepsilon(k_1, k_2, \|z\|) \|z\| + \bar{d}\rho_{f'}(k_1, \|z\|) \|z\| \\ &\leq \left(\underbrace{\rho_{\Phi}(k_1, k_2, \|z\|) + \sigma_\varepsilon(k_1, k_2, \|z\|) + \bar{d}\rho_{f'}(k_1, \|z\|)}_{\rho_0(k_1, k_2, \|z\|)} \right) \|z\| \\ &\leq \rho_0(k_1, k_2, \|z\|) \|z\| \end{aligned}$$

BOUNDING LEMMA 4.1

Recall

$$\begin{aligned} D(\kappa_d, \theta^*, \hat{\theta}, t) &\triangleq -\dot{d}(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) + \varepsilon(\kappa_d) + N_0(\kappa_d, \theta^*, \hat{\theta}) \\ &= -\dot{d}(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) + \varepsilon(\kappa_d) + \Phi(\kappa_d, \theta^*) - \Phi(\kappa_d, \hat{\theta}) \end{aligned}$$

By Bounding Lemma 2.3 and $f \in \mathcal{C}^1$ and EVT, there exists a known constant

$$\begin{aligned} \bar{D} \triangleq & \bar{d} + \bar{d} \sup_{(q_d, \dot{q}_d) \in \bar{B}_{\bar{q}_d}(0_n) \times \bar{B}_{\bar{q}_d}(0_n)} \left\| \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} \right\| \\ & + \sup_{(\kappa_d, \theta^*, \hat{\theta}) \in \bar{B}_{\bar{\kappa}_d}(0_{5n}) \times \bar{B}_{\bar{\theta}}(0_p) \times \bar{B}_{\bar{\theta}}(0_p)} \left\| \Phi(\kappa_d, \theta^*) - \Phi(\kappa_d, \hat{\theta}) \right\| + \sup_{\kappa_d \in \bar{B}_{\bar{\kappa}_d}(0_{5n})} \|\varepsilon(\kappa_d)\| \\ \left\| D(\kappa_d, \theta^*, \hat{\theta}, t) \right\| & \leq \bar{D} \end{aligned}$$

BOUNDING LEMMA 4.2

Differentiate

$$\begin{aligned} \dot{D}(\kappa_d, \theta^*, \hat{\theta}, t) &= -\ddot{d}(t) - \left(\frac{\partial f}{\partial \dot{q} \partial q} \Big|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial^2 f}{\partial \dot{q}^2} \Big|_{(q_d, \dot{q}_d)} \ddot{q}_d \right) d(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} \dot{d}(t) + \frac{d\varepsilon}{d\kappa} \Big|_{\kappa_d} \dot{\kappa}_d \\ &+ \frac{\partial \Phi}{\partial \kappa} \Big|_{(\kappa_d, \theta^*)} \dot{\kappa}_d + \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa_d, \theta^*)} \dot{\theta} - \frac{\partial \Phi}{\partial \kappa} \Big|_{(\kappa_d, \hat{\theta})} \dot{\kappa}_d - \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa_d, \hat{\theta})} \dot{\theta} \\ &= -\ddot{d}(t) - \left(\frac{\partial f}{\partial \dot{q} \partial q} \Big|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial^2 f}{\partial \dot{q}^2} \Big|_{(q_d, \dot{q}_d)} \ddot{q}_d \right) d(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} \dot{d}(t) + \frac{d\varepsilon}{d\kappa} \Big|_{\kappa_d} \dot{\kappa}_d \\ &+ \left(\frac{\partial \Phi}{\partial \kappa} \Big|_{(\kappa_d, \theta^*)} - \frac{\partial \Phi}{\partial \kappa} \Big|_{(\kappa_d, \hat{\theta})} \right) \dot{\kappa}_d - \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa_d, \hat{\theta})} \dot{\theta} \end{aligned}$$

By $\frac{d\varepsilon}{d\kappa} \in \mathcal{C}^0$, $\frac{\partial \Phi}{\partial \theta} \in \mathcal{C}^0$ ($\mathcal{C}^1$ in fact), $\frac{\partial \Phi}{\partial \kappa} \in \mathcal{C}^0$ ($\mathcal{C}^1$ in fact), EVT and Bounding Lemma 2.6, there exists an **unknown** constant

$$\begin{aligned} \bar{D} \triangleq & \bar{d} + \bar{d} \sup_{(q_d, \dot{q}_d) \in \bar{B}_{\bar{q}_d}(0_n) \times \bar{B}_{\bar{q}_d}(0_n)} \left\| \frac{\partial f}{\partial \dot{q} \partial q} \Big|_{(q_d, \dot{q}_d)} \right\| + \bar{d} \sup_{(q_d, \dot{q}_d) \in \bar{B}_{\bar{q}_d}(0_n) \times \bar{B}_{\bar{q}_d}(0_n)} \left\| \frac{\partial^2 f}{\partial \dot{q}^2} \Big|_{(q_d, \dot{q}_d)} \right\| \\ & + \bar{d} \sup_{(q_d, \dot{q}_d) \in \bar{B}_{\bar{q}_d}(0_n) \times \bar{B}_{\bar{q}_d}(0_n)} \left\| \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} \right\| + \bar{\kappa}_d \sup_{\kappa_d \leq \bar{\kappa}_d} \left\| \frac{d\varepsilon}{d\kappa} \Big|_{\kappa_d} \right\| \\ & + \bar{\kappa}_d \sup_{(\kappa_d, \theta^*, \hat{\theta}) \in \bar{B}_{\bar{\kappa}_d}(0_{5n}) \times \bar{B}_{\bar{\theta}}(0_p) \times \bar{B}_{\bar{\theta}}(0_p)} \left\| \frac{\partial \Phi}{\partial \kappa} \Big|_{(\kappa_d, \theta^*)} - \frac{\partial \Phi}{\partial \kappa} \Big|_{(\kappa_d, \hat{\theta})} \right\| \\ & + \eta_1 \sup_{(\kappa_d, \hat{\theta}) \in \bar{B}_{\bar{\kappa}_d}(0_{5n}) \times \bar{B}_{\bar{\theta}}(0_p)} \left\| \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa_d, \hat{\theta})} \right\| \end{aligned}$$

NOTE: We will return to $\frac{d\varepsilon}{d\kappa} \Big|_{\kappa_d}$ later to make it known! For now, we assume without proof that

$$\sup_{\kappa_d \leq \bar{\kappa}_d} \left\| \frac{d\varepsilon}{d\kappa} \Big|_{\kappa_d} \right\| \leq \bar{\varepsilon}_d$$

where $\bar{\varepsilon}_d$ is a **known** constant. Thus, $\bar{D}$ is now a **known** constant.

7 P-Function

Recall

$$r_2 \triangleq \dot{r}_1 + k_2 r_1$$

Introduce

$$\begin{aligned} \zeta(t) &\triangleq K_{\text{RISE}} \|r_1(t)\|_1 - r_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) \\ \phi(t) &\triangleq r_1^\top(t) \dot{D}(\kappa_d, \theta^*, \hat{\theta}, t) + (k_2 - \lambda_P) \zeta(t) \end{aligned}$$

Let

$$\dot{\eta}(t) = -\lambda_P \eta(t) + \phi(t) \quad \eta(t_0) = 0$$

Define

$$P(t) \triangleq \zeta(t) + \eta(t)$$

So,

$$\begin{aligned} \dot{P}(t) &\in \dot{\zeta}(t) + \dot{\eta}(t) \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - \dot{r}_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - r_1^\top(t) \dot{D}(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P \eta(t) + \phi(t) \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - \dot{r}_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \cancel{r_1^\top(t) \dot{D}(\kappa_d, \theta^*, \hat{\theta}, t)} - \lambda_P \eta(t) + \cancel{r_1^\top(t) \dot{D}(\kappa_d, \theta^*, \hat{\theta}, t)} + (k_2 - \lambda_P) \zeta(t) \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - \dot{r}_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P \eta(t) + (k_2 - \lambda_P) \zeta(t) \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - \dot{r}_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P \eta(t) + k_2 \zeta(t) - \lambda_P \zeta(t) \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - \dot{r}_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P (\eta(t) + \zeta(t)) + k_2 (K_{\text{RISE}} \|r_1(t)\|_1 - r_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t)) \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - \dot{r}_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P P(t) + K_{\text{RISE}} k_2 \|r_1(t)\|_1 - k_2 r_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - (\dot{r}_1^\top(t) + k_2 r_1^\top(t)) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P P(t) + K_{\text{RISE}} k_2 \|r_1(t)\|_1 \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - r_2^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P P(t) + K_{\text{RISE}} k_2 \|r_1(t)\|_1 \\ &\in K_{\text{RISE}} \dot{r}_1^\top(t) K[\text{sgn}(r_1(t)) - r_2^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P P(t) + K_{\text{RISE}} k_2 r_1^\top(t) K[\text{sgn}(r_1(t)) \\ &\in K_{\text{RISE}} (\dot{r}_1^\top(t) + k_2 r_1^\top(t)) K[\text{sgn}(r_1(t)) - r_2^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P P(t) \\ &\in K_{\text{RISE}} r_2^\top(t) K[\text{sgn}(r_1(t)) - r_2^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P P(t) \\ &\in -\lambda_P P(t) + r_2^\top(t) (K_{\text{RISE}} K[\text{sgn}(r_1(t)) - D(\kappa_d, \theta^*, \hat{\theta}, t)]) \end{aligned}$$

NOTE: $P(t_0)$ must satisfy:

$$\begin{aligned} P(t_0) &= \zeta(t_0) \\ &= K_{\text{RISE}} \|r_1(t_0)\|_1 - r_1^\top(t_0) D(\kappa_d(t_0), \theta^*, \hat{\theta}(t_0), t_0) \end{aligned}$$

Define

$$P_0 \triangleq K_{\text{RISE}} \|r_1(t_0)\|_1 - r_1^\top(t_0) D(\kappa_d(t_0), \theta^*, \hat{\theta}(t_0), t_0)$$

NOTE:

$$\begin{aligned} |P(t_0)| &\leq K_{\text{RISE}} \|r_1(t_0)\|_1 + \|r_1(t_0)\| \overline{D} \\ &\leq \sqrt{n} K_{\text{RISE}} \|r_1(t_0)\| + \|r_1(t_0)\| \overline{D} \\ &\leq \left(\underbrace{\sqrt{n} K_{\text{RISE}} + \overline{D}}_{C_P} \right) \|r_1(t_0)\| \\ &\leq C_P \|r_1(t_0)\| \\ &\leq C_P \|z(t_0)\| \end{aligned}$$

How do we know $P \geq 0$ for all time? Use sufficient conditions. An obvious first step is to ensure $\|\zeta(t)\| \geq 0$ for all time. Since $\|r\|_1 \geq \|r\|$,

$$\begin{aligned} K_{\text{RISE}} \|r_1(t)\|_1 - r_1^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t) &\geq K_{\text{RISE}} \|r_1(t)\|_1 - \|r_1(t)\| \overline{D} \\ &\geq K_{\text{RISE}} \|r_1(t)\| - \overline{D} \|r_1(t)\| \\ &\geq \left(\underbrace{K_{\text{RISE}} - \overline{D}}_{b_1} \right) \|r_1(t)\| \end{aligned}$$

To get $b_1 > 0$, select (choice of strict inequality will be apparent later)

$$K_{\text{RISE}} > \bar{D}$$

and thus

$$\|\zeta(t)\| \geq b_1 \|r_1(t)\| > 0$$

Let $\lambda_P \in (0, k_2)$ and $b_2 \triangleq k_2 - \lambda_P$. Thus $b_2 > 0$. Next, we want $\phi(t) \geq 0$.

$$\begin{aligned} r_1^\top(t) \dot{D}(\kappa_d, \theta^*, \hat{\theta}, t) + (k_2 - \lambda_P) \zeta(t) &= r_1^\top(t) \dot{D}(\kappa_d, \theta^*, \hat{\theta}, t) + b_2 \zeta(t) \\ &> -\|r_1(t)\| \bar{\bar{D}} + b_1 b_2 \|r_1(t)\| \\ &> (b_1 b_2 - \bar{\bar{D}}) \|r_1(t)\| \end{aligned}$$

We need

$$\begin{aligned} b_1 b_2 - \bar{\bar{D}} &\geq 0 \\ b_1 b_2 &\geq \bar{\bar{D}} \\ b_1 &\geq \frac{\bar{\bar{D}}}{k_2 - \lambda_P} \\ K_{\text{RISE}} - \bar{D} &\geq \frac{\bar{\bar{D}}}{k_2 - \lambda_P} \\ K_{\text{RISE}} &\geq \bar{D} + \frac{\bar{\bar{D}}}{k_2 - \lambda_P} \end{aligned}$$

So the only gain suggestion you need (well, it's sufficient)

$$K_{\text{RISE}} \geq \bar{D} + \frac{\bar{\bar{D}}}{k_2 - \lambda_P}$$

NOTE: The above requires

$$\lambda_P \in (0, k_2)$$

which can always be satisfied since $k_2 > 0$. However, we will need to mention this in the gain conditions. Note that the solution to the $\eta(t)$ ODE is

$$\begin{aligned} \eta(t) &= \eta(t_0) e^{-\lambda_P(t-t_0)} + \int_{t_0}^t e^{-\lambda_P(t-\tau)} \phi(\tau) d\tau \\ &= \int_{t_0}^t e^{-\lambda_P(t-\tau)} \phi(\tau) d\tau \end{aligned}$$

Clearly a sufficient condition for $\eta(t) \geq 0$ for all t is that $\phi(t) \geq 0$ for all t , which was shown. Thus, $\eta(t) \geq 0$ for all t .

8 Stability

Let

$$z_P \triangleq \begin{bmatrix} z^\top & P \end{bmatrix}^\top \in \mathbb{R}^{3n+1}$$

Consider the differential inclusion

$$z_P \in H(z_P, t)$$

$$H(z_P, t) \triangleq \left\{ \begin{bmatrix} r_1 - k_1 e \\ r_2 - k_2 r_1 \\ \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - k_3 r_2 - r_1 - K_{\text{RISE}} \nu_1 + D(\kappa_d, \theta^*, \hat{\theta}, t) \\ -\lambda_P P(t) + r_2^\top(t) (K_{\text{RISE}} \nu_2 - D(\kappa_d, \theta^*, \hat{\theta}, t)) \end{bmatrix} : \nu_1, \nu_2 \in K[\text{sgn}(r_1(t))] \right\}$$

Stability. Consider the Lyapunov candidate

$$V(z_P) = \frac{1}{2} z^\top z + P$$

Trajectory derivative

$$\begin{aligned} \dot{V}(z_P, t) &= e^\top (r_1 - k_1 e) + r_1^\top (r_2 - k_2 r_1) + r_2^\top \left(\tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - k_3 r_2 - r_1 - K_{\text{RISE}} \nu_1 + D(\kappa_d, \theta^*, \hat{\theta}, t) \right) \\ &\quad - \lambda_P P + r_2^\top(t) \left(K_{\text{RISE}} \nu_2 - D(\kappa_d, \theta^*, \hat{\theta}, t) \right) \\ &= \cancel{e^\top r_1} - k_1 e^\top e + \cancel{r_1^\top r_2} - k_2 r_1^\top r_1 + r_2^\top \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - k_3 r_2^\top r_2 - \cancel{r_2^\top r_1} - K_{\text{RISE}} r_2^\top \nu_1 + \cancel{r_2^\top D(\kappa_d, \theta^*, \hat{\theta}, t)} \\ &\quad - \lambda_P P + K_{\text{RISE}} r_2^\top(t) \nu_2 - \cancel{r_2^\top(t) D(\kappa_d, \theta^*, \hat{\theta}, t)} \\ &= -k_1 \|e\|^2 - k_2 \|r_1\|^2 - k_3 \|r_2\|^2 + r_2^\top \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) + K_{\text{RISE}} r_2^\top (\nu_2 - \nu_1) - \lambda_P P + \|e\| \|r_1\| \\ &\stackrel{\text{a.e.}}{=} -k_1 \|e\|^2 - k_2 \|r_1\|^2 - k_3 \|r_2\|^2 + r_2^\top \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P P + \frac{1}{2} \|e\|^2 + \frac{1}{2} \|r_1\|^2 \\ &\stackrel{\text{a.e.}}{=} -\left(k_1 - \frac{1}{2}\right) \|e\|^2 - \left(k_2 - \frac{1}{2}\right) \|r_1\|^2 - k_3 \|r_2\|^2 + r_2^\top \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - \lambda_P P \end{aligned}$$

Using Bounding Lemma 3.4 and triangle inequality,

$$\begin{aligned} \dot{V}(z_P) &\stackrel{\text{a.e.}}{\leq} -\left(k_1 - \frac{1}{2}\right) \|e\|^2 - \left(k_2 - \frac{1}{2}\right) \|r_1\|^2 + \|r_2\| \rho_0(k_1, k_2, \|z\|) \|z\| - k_3 \|r_2\|^2 - \lambda_P P \\ &\stackrel{\text{a.e.}}{\leq} -\left(k_1 - \frac{1}{2}\right) \|e\|^2 - \left(k_2 - \frac{1}{2}\right) \|r_1\|^2 + \|r_2\| \rho_0(k_1, k_2, \|z\|) (\|e\| + \|r_1\| + \|r_2\|) - k_3 \|r_2\|^2 - \lambda_P P \\ &\stackrel{\text{a.e.}}{\leq} -\left(k_1 - \frac{1}{2}\right) \|e\|^2 - \left(k_2 - \frac{1}{2}\right) \|r_1\|^2 \\ &\quad + \rho_0(k_1, k_2, \|z\|) \|e\| \|r_2\| + \rho_0(k_1, k_2, \|z\|) \|r_1\| \|r_2\| + \rho_0(k_1, k_2, \|z\|) \|r_2\|^2 - k_3 \|r_2\|^2 - \lambda_P P \\ &\stackrel{\text{a.e.}}{\leq} -\left(k_1 - \frac{1}{2}\right) \|e\|^2 - \left(k_2 - \frac{1}{2}\right) \|r_1\|^2 + \frac{1}{2} \|e\|^2 + \frac{1}{2} \rho_0^2(k_1, k_2, \|z\|) \|r_2\|^2 + \frac{1}{2} \|r_1\|^2 \\ &\quad + \frac{1}{2} \rho_0^2(k_1, k_2, \|z\|) \|r_2\|^2 + \rho_0(k_1, k_2, \|z\|) \|r_2\|^2 - k_3 \|r_2\|^2 - \lambda_P P \\ &\stackrel{\text{a.e.}}{\leq} -(k_1 - 1) \|e\|^2 - (k_2 - 1) \|r_1\|^2 \\ &\quad + \frac{1}{2} \rho_0^2(k_1, k_2, \|z\|) \|r_2\|^2 + \frac{1}{2} \rho_0^2(k_1, k_2, \|z\|) \|r_2\|^2 + \rho_0(k_1, k_2, \|z\|) \|r_2\|^2 - k_3 \|r_2\|^2 - \lambda_P P \\ &\stackrel{\text{a.e.}}{\leq} -(k_1 - 1) \|e\|^2 - (k_2 - 1) \|r_1\|^2 - \left(k_3 - \underbrace{\left(\rho_0^2(k_1, k_2, \|z\|) + \rho_0(k_1, k_2, \|z\|) \right)}_{\rho(k_1, k_2, \|z\|)} \right) \|r_2\|^2 - \lambda_P P \\ &\stackrel{\text{a.e.}}{\leq} -(k_1 - 1) \|e\|^2 - (k_2 - 1) \|r_1\|^2 - (k_3 - \rho(k_1, k_2, \|z\|)) \|r_2\|^2 - \lambda_P P \end{aligned}$$

Pick $\Delta, \lambda_V \in \mathbb{R}_{>0}$. Let the desired region of attraction (for z not z_P !) be $\mathcal{S} \triangleq \{\zeta \in \mathbb{R}^{5n} : \|\zeta\| \leq \Delta\}$. The gain conditions should be selected to ensure that for all $z \in \mathcal{S}$,

$$\dot{V}(z_P) \stackrel{\text{a.e.}}{\leq} -\frac{\lambda_V}{2} \|z\|^2 - \lambda_P P$$

So,

$$-(k_1 - 1) \|e\|^2 - (k_2 - 1) \|r_1\|^2 - (k_3 - \rho(k_1, k_2, \|z\|)) \|r_2\|^2 - \lambda_P P \stackrel{\text{a.e.}}{\leq} -\min\{k_1 - 1, k_2 - 1, k_3 - \rho(k_1, k_2, \Delta)\} \|z\|^2 - \lambda_P P$$

Compare

$$\begin{aligned} -\min \{k_1 - 1, k_2 - 1, k_3 - \rho(k_1, k_2, \Delta)\} \|z\|^2 &\stackrel{\text{a.e.}}{\leq} -\frac{\lambda_V}{2} \|z\|^2 \\ \min \{k_1 - 1, k_2 - 1, k_3 - \rho(k_1, k_2, \Delta)\} &\stackrel{\text{a.e.}}{\leq} \frac{\lambda_V}{2} \end{aligned}$$

Thus,

$$\begin{aligned} \dot{V}(z_P) &\stackrel{\text{a.e.}}{\leq} -\frac{\lambda_V}{2} \|z\|^2 - \lambda_P P \\ &\stackrel{\text{a.e.}}{\leq} -\min \{\lambda_V, \lambda_P\} \underbrace{\left(\frac{1}{2} \|z\|^2 + P \right)}_{V(z_P)} \end{aligned}$$

Let

$$\lambda \triangleq \frac{1}{2} \min \{\lambda_V, \lambda_P\}$$

Thus,

$$\dot{V}(z_P) \stackrel{\text{a.e.}}{\leq} -2\lambda (V(z_P))$$

9 GAIN CONDITIONS

Let λ be a desired convergence rate and let Δ be the size of desired initial conditions ($\|z(t_0)\| \leq \Delta$). Let

$$C_\Delta \triangleq \sqrt{\Delta^2 + 2C_P\Delta}$$

The sufficient gain conditions are

$$\begin{aligned} \lambda_V &\geq 2\lambda \\ \lambda_P &\geq 2\lambda \\ k_1 &> 1 + \frac{\lambda_V}{2} \\ k_2 &> \max \left\{ 1 + \frac{\lambda_V}{2}, \lambda_P \right\} \\ k_3 &> \rho(k_1, k_2, C_\Delta) + \frac{\lambda_V}{2} \\ K_{\text{RISE}} &\geq \bar{D} + \frac{\bar{D}}{k_2 - \lambda_P} \end{aligned}$$

Interestingly intuition: how “bad” the disturbances and weight initialization are will affect the y-intercept of the convergence envelope, where as the gains themselves drive the time constant of that convergence envelope. Recall

$$V(z_P) = \frac{1}{2} z^\top z + P$$

Since $P \geq 0$,

$$\frac{1}{2} \|z(t)\|^2 \leq V(z_P(t))$$

Begin with the premise that

$$\|z(t_0)\| \leq \Delta$$

Since $|P(t_0)| \leq C_P \|z(t_0)\|$,

$$\begin{aligned} V(z_P(t_0)) &\leq \frac{1}{2} \|z(t_0)\|^2 + C_P \|z(t_0)\| \\ &\leq \frac{1}{2} \Delta^2 + C_P \Delta \end{aligned}$$

Define the sublevel set

$$\Omega_S \triangleq \left\{ z_P \in \mathbb{R}^{3n+1} : V(z_P(t)) \leq \frac{1}{2} \Delta^2 + C_P \Delta \right\}$$

Thus $\|z(t_0)\| \leq \Delta$ implies that $z_P(t_0) \in \Omega_S$. Pick some $z_P(t) \in \Omega_S$. Since $\frac{1}{2}\|z(t)\|^2 \leq V(z_P(t))$, it follows that for this $z_P(t)$ that

$$\begin{aligned}\frac{1}{2}\|z(t)\|^2 &\leq \frac{1}{2}\Delta^2 + C_P\Delta \\ \|z(t)\|^2 &\leq \Delta^2 + 2C_P\Delta \\ \|z(t)\| &\leq \underbrace{\sqrt{\Delta^2 + 2C_P\Delta}}_{C_\Delta}\end{aligned}$$

Consider the set of initial states

$$\mathcal{S}_P \triangleq \{z_P \in \mathbb{R}^{3n+1} : \|z\| \leq C_\Delta\}$$

Recall that P is not independent of z , hence why P is not specified independently above. Thus $z_P \in \Omega_S \implies z_P \in \mathcal{S}_P$ and it follows that

$$\Omega_S \subseteq \mathcal{S}_P$$

Since $z_P(t_0) \in \Omega_S$ and $\Omega_S \subseteq \mathcal{S}_P$, our gain conditions guarantee that for all $z_P \in \mathcal{S}_P$,

$$\dot{V}(z_P(t)) \underset{\text{a.e.}}{\leq} -2\lambda(V(z_P(t)))$$

It was shown that the above implies

$$\begin{aligned}V(z_P(t)) &\underset{\text{a.e.}}{\leq} V(z_P(t_0))e^{-2\lambda(t-t_0)} \\ &\underset{\text{a.e.}}{\leq} V(z_P(t_0)) \\ &\underset{\text{a.e.}}{\leq} \frac{1}{2}\Delta^2 + C_P\Delta\end{aligned}$$

Thus, Ω_S is forward invariant. Solving the differential inequality...(Really just invoke Comparison Lemma or Gronwall inequality)

$$\begin{aligned}s\mathcal{L}\{V(z_P(t))\} - V(z_P(t_0)) &\underset{\text{a.e.}}{\leq} -2\lambda\mathcal{L}\{V(z(t))\} \\ (s+2\lambda)\mathcal{L}\{V(z_P(t))\} &\underset{\text{a.e.}}{\leq} V(z_P(t_0)) \\ \mathcal{L}\{V(z_P(t))\} &\underset{\text{a.e.}}{\leq} \frac{V(z_P(t_0))}{s+2\lambda} \\ \mathcal{L}^{-1}\{\mathcal{L}\{V(z_P(t))\}\} &\underset{\text{a.e.}}{\leq} \mathcal{L}^{-1}\left\{\frac{V(z_P(t_0))}{s+2\lambda}\right\} \\ V(z_P(t)) &\underset{\text{a.e.}}{\leq} V(z_P(t_0))e^{-2\lambda(t-t_0)}\end{aligned}$$

Since

$$\begin{aligned}V(z_P(t_0)) &= \frac{1}{2}z^\top(t_0)z(t_0) + P(t_0) \\ V(z_P(t)) &= \frac{1}{2}z^\top(t)z(t) + P(t)\end{aligned}$$

and

$$|P(t_0)| \leq C_P\|z(t_0)\|$$

then, it follows that

$$\begin{aligned}
\frac{1}{2} z^\top(t) z(t) + P(t) &\leq_{\text{a.e.}} \left(\frac{1}{2} z^\top(t_0) z(t_0) + P(t_0) \right) e^{-2\lambda(t-t_0)} \\
\frac{1}{2} \|z(t)\|^2 &\leq_{\text{a.e.}} \left(\frac{1}{2} \|z(t_0)\|^2 + C_P \|z(t_0)\| \right) e^{-2\lambda(t-t_0)} \\
\frac{1}{2} \|z(t)\|^2 &\leq_{\text{a.e.}} \left(\frac{1}{2} \|z(t_0)\|^2 + C_P \|z(t_0)\| \right) e^{-2\lambda(t-t_0)} \\
\|z(t)\|^2 &\leq_{\text{a.e.}} \left(\|z(t_0)\|^2 + 2C_P \|z(t_0)\| \right) e^{-2\lambda(t-t_0)} \\
\|z(t)\| &\leq_{\text{a.e.}} \sqrt{\left(\|z(t_0)\|^2 + 2C_P \|z(t_0)\| \right) e^{-2\lambda(t-t_0)}} \\
\|z(t)\| &\leq_{\text{a.e.}} \underbrace{\sqrt{\left(\|z(t_0)\|^2 + 2C_P \|z(t_0)\| \right) e^{-\lambda(t-t_0)}}}_{\substack{\alpha \in \mathcal{K}_\infty \\ \beta \in \mathcal{KL}_\infty}}
\end{aligned}$$

Thus the differential inclusion is uniformly semi-globally asymptotically stable (USGAS) in the strong sense. States converge exponentially but since α is not linear in $\|z(t_0)\|$, the system is not USGES!

10 BOUNDEDNESS

1. Stability analysis showed that $V \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^{3n+1})$. Thus, $e, r_1, r_2, \dot{e}, \ddot{e} \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$, $P \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R})$, and $z \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^{3n})$.
2. Since $q_d \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$ by formulation and $e \triangleq q_d - q$ and $e \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$, it follows that $q \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$.
3. Since $\dot{q}_d \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$ by formulation and $r_1 = \dot{q}_d - \dot{q} + k_1 e$ and $e \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$, it follows that $\dot{q} \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$.
4. Since $\ddot{q}_d \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$ by formulation and $r_2 = \ddot{q}_d - \ddot{q} + k_1 \dot{e} + k_2 r_1$ and $r_1, \dot{e} \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$, it follows that $\ddot{q} \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$.
5. Since $z \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^{3n})$, $\tilde{u} \triangleq u_d - u$, $\|\tilde{u}\| \leq \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\|$ (Bounding Lemma 2.4), then triangle inequality implies that $u \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$ because $\|u\| \leq \bar{u}_d + \|\tilde{u}\|$ (Bounding Lemma 2.1 for $\bar{u}_d$ bound).
6. Since $q, \dot{q}, q_d, \dot{q}_d, u \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$, it follows that $\kappa \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^{5n})$.
7. Since $\hat{\theta} \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^p)$ by formulation and $\kappa \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^{5n})$ and $\Phi \in \mathcal{C}^0$ ($\mathcal{C}^2$ in fact), then by EVT, $\Phi \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$.
8. Recall $\dot{u} = (k_1 + k_2 + k_3) r_2 + (1 - k_1^2 - k_2^2 - k_1 k_2) r_1 + k_1^3 e + K_{\text{RISES}} \text{sgn}(r_1) + \Phi(\kappa, \hat{\theta})$. Since $\hat{\theta} \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^p)$ by formulation and $e, r_1, r_2 \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$ and $\text{sgn}(r_1(t)) \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$ trivially and $\kappa \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^{5n})$, it follows that $\dot{u} \in \mathcal{L}_\infty(\mathbb{R}_{\geq t_0}; \mathbb{R}^n)$.

11 FUNCTION APPROXIMATION

DOMAIN OF APPROXIMATION

Recall

$$\kappa \triangleq \begin{bmatrix} q^\top & \dot{q}^\top & q_d^\top & \dot{q}_d^\top & u^\top \end{bmatrix}^\top \in \mathbb{R}^{5n}$$

By Bounding Lemma 2.4,

$$\begin{aligned}
\|\kappa\| &\leq \|q\| + \|\dot{q}\| + \|q_d\| + \|\dot{q}_d\| + \|u\| \\
&\leq \|q_d - e\| + \|\dot{q}_d - \dot{e}\| + \|q_d\| + \|\dot{q}_d\| + \|u\| \\
&\leq \bar{q}_d + \|e\| + \bar{\dot{q}}_d + \|\dot{e}\| + \bar{q}_d + \bar{\dot{q}}_d + \bar{u}_d + \|\tilde{u}\| \\
&\leq 2\bar{q}_d + 2\bar{\dot{q}}_d + \|e\| + \|r_1 - k_1 e\| + \bar{u}_d + \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\| \\
&\leq 2\bar{q}_d + 2\bar{\dot{q}}_d + \|e\| + \|r_1\| + k_1 \|e\| + \bar{u}_d + \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\| \\
&\leq 2\bar{q}_d + 2\bar{\dot{q}}_d + \|z\| + \|z\| + k_1 \|z\| + \bar{u}_d + \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\| \\
&\leq 2\bar{q}_d + 2\bar{\dot{q}}_d + (k_1 + 2) \|z\| + \bar{u}_d + \rho_{\tilde{u}}(k_1, k_2, \|z\|) \|z\|
\end{aligned}$$

Since stability showed that $\|z\| \leq C_\Delta$ (not $\Delta!$), thus,

$$\begin{aligned}\|\kappa\| &\leq \bar{\kappa} \\ \bar{\kappa} &\triangleq 2\bar{q}_d + 2\bar{\dot{q}}_d + (k_1 + 2) C_\Delta + \bar{u}_d + \rho_{\bar{u}}(k_1, k_2, \Delta) C_\Delta\end{aligned}$$

This will be just one of three candidate radii for our approximation set! Let a desired accuracy be $\bar{\varepsilon}_d \in \mathbb{R}_{>0}$. In Bounding Lemma 3.4, we needed the fact that

$$\sup_{\kappa_d \leq \bar{\kappa}_d} \left\| \frac{d\varepsilon}{d\kappa} \Big|_{\kappa_d} \right\| \leq \bar{\varepsilon}_d$$

Recall (by Bounding Lemma 2.1 and 2.3)

$$\bar{\kappa}_d \triangleq 2\bar{q}_d + 2\bar{\dot{q}}_d + \bar{u}_d$$

Additionally, to make the bound in Bounding Lemma 3.1 **known** rather than **unknown**, we need

$$\sup_{z \in \mathcal{S}} \left(\sup_{\kappa \in \bar{B}_{\bar{\kappa}_d + \rho_{\bar{\kappa}}(k_1, k_2, \|z\|)}(\mathbf{0}_n)} \left(\left\| \frac{d\varepsilon}{d\kappa} \Big|_{\kappa} \right\| \right) \right) \leq \bar{\varepsilon}_d$$

Recall

$$\begin{aligned}\sigma_\varepsilon(k_1, k_2, \|z\|) &= \sup_{\kappa \in \bar{B}_{\bar{\kappa}_d + \rho_{\bar{\kappa}}(k_1, k_2, \|z\|)}(\mathbf{0}_n)} \left(\left\| \frac{d\varepsilon}{d\kappa} \Big|_{\kappa} \right\| \right) \rho_{\bar{\kappa}}(k_1, k_2, \|z\|) \\ \sigma_\varepsilon(k_1, k_2, \|z\|) &= \sup_{\kappa \in \bar{B}_{\bar{\kappa}_d + (k_1 + 2 + \rho_{\bar{u}}(k_1, k_2, \|z\|))\|z\|}(\mathbf{0}_n)} \left(\left\| \frac{d\varepsilon}{d\kappa} \Big|_{\kappa} \right\| \right) \rho_{\bar{\kappa}}(k_1, k_2, \|z\|)\end{aligned}$$

Thus, the three candidate radii for our approximation set are:

$$\begin{aligned}R_\Omega^{\text{Candidate}} &\triangleq \bar{\kappa}_d + (k_1 + 2) C_\Delta + \rho_{\bar{u}}(k_1, k_2, C_\Delta) C_\Delta \\ \bar{\kappa}_d &\triangleq 2\bar{q}_d + 2\bar{\dot{q}}_d + \bar{u}_d \\ \bar{\kappa} &\triangleq \underbrace{2\bar{q}_d + 2\bar{\dot{q}}_d + \bar{u}_d}_{\bar{\kappa}_d} + (k_1 + 2) C_\Delta + \rho_{\bar{u}}(k_1, k_2, C_\Delta) C_\Delta\end{aligned}$$

To ensure the same approximation accuracy holds in all three cases, let the domain for function approximation be

$$\begin{aligned}\Omega &\triangleq \{\zeta \in \mathbb{R}^{5n} : \|\zeta\| \leq \bar{\kappa}\}. \\ \bar{\kappa} &\triangleq \bar{\kappa}_d + (k_1 + 2) C_\Delta + \rho_{\bar{u}}(k_1, k_2, C_\Delta) C_\Delta\end{aligned}$$

Since $\rho_{\bar{u}}$ and $\bar{u}_d$ are assumed known a priori (and only depends on our user-defined Δ), we retroactively proclaim Ω to be our domain of approximation.

WHAT CAN WE APPROXIMATE?

Recall

$$F(\kappa) \triangleq \frac{\partial h}{\partial q_d} \Big|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial h}{\partial \dot{q}_d} \Big|_{(q_d, \dot{q}_d)} h(q_d, \dot{q}_d) - \frac{\partial f}{\partial q} \Big|_{(q, \dot{q})} \dot{q} - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} f(q, \dot{q}) - \frac{\partial f}{\partial \ddot{q}} \Big|_{(q, \dot{q})} u$$

A deep residual neural network will approximate $F(\kappa)$ as $\Phi(\kappa, \hat{\theta})$. Here, $L_{\text{in}} = 5n$ and $L_{\text{out}} = n$. The activation functions $\varsigma_{i,j} : \mathbb{R} \rightarrow \mathbb{R}$ must be chosen to be of class $\mathcal{A}_{\text{quad}}$.

DEFINITION [?, Definition 3.1]

The activation function $\varsigma : \mathbb{R} \rightarrow \mathbb{R}$ is said to be of class $\mathcal{A}_{\text{quad}}$ (denoted $\varsigma \in \mathcal{A}_{\text{quad}}$) if it is (i) nondecreasing, (ii) non-constant, and (iii) there exists an $\ell \in \mathbb{Z}_{\geq 0}$ such that ς is a globally-defined solution to $\frac{d^\ell \varsigma}{ds} = a_0 + a_1 \varsigma + a_2 \varsigma^2$, where $\varsigma \triangleq \frac{d^{(\ell)} \varsigma}{ds^{(\ell)}}$ is injective, $a_0, a_1, a_2 \in \mathbb{R}$, $a_2 \neq 0$, and $\frac{d^{(0)} \varsigma}{ds^{(0)}} = \varsigma$. By consequence of Definition 11, any $\varsigma \in \mathcal{A}_{\text{quad}}$ is necessarily an analytical function. Thus, ς and all its derivatives are locally Lipschitz on $\mathbb{R}$. Hyperbolic tangent, sigmoid, and softplus are of class $\mathcal{A}_{\text{quad}}$ but ReLU, swish, and the identity mapping are not. The function $\varsigma(s) = \frac{1}{c} \log(1 + e^{cs})$ approximates ReLU for large $c \in \mathbb{R}_{>0}$ and is of class $\mathcal{A}_{\text{quad}}$. Leaky ReLU is given as $\varsigma(s) = s$ for all $s \in \mathbb{R}_{\geq 0}$ and $\varsigma(s) = rs$ (with $r \in \mathbb{R}_{>0}$) for all $s < 0$. The function $\varsigma(s) = rs + \frac{1}{c} \log(1 + e^{(1-r)cs})$ approximates leaky ReLU for large $c \in \mathbb{R}_{>0}$ and is of class $\mathcal{A}_{\text{quad}}$.

The universal approximation theorem for ResNets is given as follows.

LEMMA [?, Corollary 5.2]

(Universal Approximation Theorem for Residual Neural Networks). For any compact domain $\Omega \subseteq \mathbb{R}^{L_{\text{in}}}$, continuous function $F : \Omega \rightarrow \mathbb{R}^{L_{\text{out}}}$, and approximation accuracy $\bar{\varepsilon} \in \mathbb{R}_{>0}$, there exists a number of blocks $b \in \mathbb{Z}_{>0}$ (thus, a $\mathbf{p} \in \mathbb{Z}_{>0}$) and a parameter vector $\theta^* \in \mathbb{R}^{\mathbf{p}}$ such that for a ResNet architecture $\Psi : \mathbb{R}^{L_{\text{in}}} \times \mathbb{R}^{\mathbf{p}} \rightarrow \mathbb{R}^{L_{\text{out}}}$ with blocks of width $w = \max(L_{\text{in}}, L_{\text{out}}) + 1$ (i.e., $L_{i,j} \geq w$ for all $(i, j) \in \{0, \dots, b\} \times \{1, \dots, k_i\}$) and depth 1 (i.e., $k_i \geq 1$ for all $i \in \{0, \dots, b\}$),

$$\sup_{x \in \Omega} \|\varepsilon(x)\| \leq \bar{\varepsilon}, \quad (2)$$

where $\varepsilon(x) \triangleq h(x) - \Psi(x, \theta^*)$, provided that for all $(i, j) \in \{0, \dots, b\} \times \{0, \dots, k_i\}$, the activation functions $\varsigma_{i,j} \in \mathcal{A}_{\text{quad}}$.

Here, note that $L_{\text{out}} = n$ and $L_{\text{in}} = 5n$. So, w should theoretically be at least $5n + 1$.

By formulation, F is $\mathcal{C}^0$ on $\mathbb{R}^n$ ($\mathcal{C}^1$ in fact). Thus, for any given compact set $\Omega \subseteq \mathbb{R}^n$, Lemma 11 applies. However, notice that the θ^* resulting from a desired approximation set and desired approximation accuracy lies in $\mathbb{R}^{\mathbf{p}}$. Surely, for any fixed F , if there exists a θ^* such that (2) holds, there exists a $\bar{\theta}$ such that $\|\theta^*\| \leq \bar{\theta}$. However, there is no means of determining this $\bar{\theta}$. Knowing it fundamentally conflicts with the underlying challenge of nonlinear regression anyways! Conversely, in practice, $\bar{\theta}$ is specified a priori. This results in a dilemma. The user is forced to pick two of the following three: (1) F can be any $\mathcal{C}^0$ function, (2) F can be approximated arbitrarily well ($\bar{\varepsilon} \rightarrow 0$), (3) a known bound exists on the weights needed to achieve the desired accuracy for the target function. Recall that ρ must be known to verify the earlier gain conditions since it contains $\bar{\varepsilon}$. Thus, (2) is required. Practical use of the developed controller dictates (3) be required. Thus, F may not be ANY continuous function. We are forced to the following assumption to resolve the dilemma. Fix a Φ (i.e., a number of blocks, width per block, and depth) and a desired accuracy $\bar{\varepsilon}_d \in \mathbb{R}_{>0}$.

ASSUMPTION.

The function F belongs to a strict subset of all $\mathcal{C}^1$ functions: those whose outputs and first derivatives can be approximated arbitrarily well with bounded weights by the chosen ResNet structure. Let $\|\varepsilon\|_{1,\Omega} \triangleq \max_{x \in \Omega} \|\Phi(x, \theta^*) - F(x)\| + \max_{x \in \Omega} \left\| \frac{\partial \Phi}{\partial x} \Big|_{(x, \theta^*)} - \frac{dF}{dx} \Big|_x \right\|$. It is assumed that $F \in \mathcal{F} \triangleq \left\{ \mathcal{F} \in \mathcal{C}^1(\Omega; \mathbb{R}^n) : \exists \theta^* \in \mathcal{U} : \|\varepsilon\|_{1,\Omega} \leq \bar{\varepsilon}_d \right\}$.

In practice, one needs to iterate with both Φ and $\bar{\theta}$ offline on similar systems to the desired application to ensure the above assumption is met, albeit heuristically. The chosen adaptive update law and initial weights must satisfy an additional assumption.

ASSUMPTION.

The weight update law $\dot{\hat{\theta}}$ must be selected to ensure that (i) $\|\dot{\hat{\theta}}\| \leq \eta_1$, (ii) $\|\hat{\theta}\| \leq \bar{\theta}$ and (iii) the initial weights and update law must be chosen such that $\|\varepsilon\|_{1,\Omega} \leq \bar{\varepsilon}_d$ at all times. That is, we must start with a good enough approximation of F over Ω and the subsequent update law must not “forget” this initial knowledge, however poor it may be (e.g., $\bar{\varepsilon}_d \gg 0$).

In practice, a lack of pre-training (and memory to prevent this “forgetting”) will not be necessary to see good performance. The controller will still work well if other things are tuned well.

12 Dual Analysis to Design Update Law

NOTE: The following section will not appear in the paper. It is just to make sure the sign in front of $\frac{\partial \hat{\Phi}}{\partial \theta}$ in the update law is correct!

First

$$\ddot{q}_d = h(q_d, \dot{q}_d)$$

Second,

$$\ddot{q} = f(q, \dot{q}) + u + d(t)$$

where $f, h \in \mathcal{C}^2$. Assume $q, \dot{q}, q_d$, and $\dot{q}_d$ are known, $d \in \mathcal{C}^2$ is unknown, and $q_d \in \mathcal{C}^3$, where

$$\begin{aligned} \|d(t)\| &\leq \bar{d} \\ \|\dot{d}(t)\| &\leq \bar{\dot{d}} \\ \|\ddot{d}(t)\| &\leq \bar{\ddot{d}} \end{aligned}$$

are known constants. Assume

$$\begin{aligned}\|q_d(t)\| &\leq \bar{q}_d \\ \|\dot{q}_d(t)\| &\leq \bar{\dot{q}}_d \\ \|\ddot{q}_d(t)\| &\leq \bar{\ddot{q}}_d \\ \|\ddot{\dot{q}}_d(t)\| &\leq \bar{\ddot{\dot{q}}}_d\end{aligned}$$

are known constants. Define

$$\begin{aligned}e &\triangleq q_d - q \\ r_1 &\triangleq \dot{e} + k_1 e \\ r_2 &\triangleq \dot{r}_1 + k_2 r_1\end{aligned}$$

Assume

$$\begin{aligned}\|\hat{\theta}\| &\leq \bar{\theta} \\ \|\theta^*\| &\leq \bar{\theta}\end{aligned}$$

13 Error Dynamics

E Error derivative

$$\dot{e} = r_1 - k_1 e$$

R1 Error derivative

$$\dot{r}_1 = r_2 - k_2 r_1$$

R2 Error derivative

1. First:

$$\begin{aligned}\dot{r}_1 &= \ddot{e} + k_1 \dot{e} \\ &= \ddot{q}_d - \ddot{q} + k_1 (r_1 - k_1 e) \\ &= h(q_d, \dot{q}_d) - (f(q, \dot{q}) + u + d(t)) + k_1 r_1 - k_1^2 e \\ &= h(q_d, \dot{q}_d) - f(q, \dot{q}) - u - d(t) + k_1 r_1 - k_1^2 e\end{aligned}$$

2. Second:

$$\begin{aligned}
\dot{r}_2 &= \ddot{r}_1 + k_2 \dot{r}_1 \\
&= \frac{d}{dt} (h(q_d, \dot{q}_d) - f(q, \dot{q}) - u - d(t) + k_1 r_1 - k_1^2 e) + k_2 \dot{r}_1 \\
&= \frac{\partial h}{\partial q_d} \bigg|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial h}{\partial \dot{q}_d} \bigg|_{(q_d, \dot{q}_d)} \ddot{q}_d - \frac{\partial f}{\partial q} \bigg|_{(q, \dot{q})} \dot{q} - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} \ddot{q} - \dot{u} - \dot{d}(t) + k_1 \dot{r}_1 - k_1^2 \dot{e} + k_2 \dot{r}_1 \\
&= \frac{\partial h}{\partial q_d} \bigg|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial h}{\partial \dot{q}_d} \bigg|_{(q_d, \dot{q}_d)} h(q_d, \dot{q}_d) - \frac{\partial f}{\partial q} \bigg|_{(q, \dot{q})} \dot{q} - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} (f(q, \dot{q}) + u + d(t)) \\
&\quad - \dot{u} - \dot{d}(t) + k_1 \dot{r}_1 - k_1^2 (r_1 - k_1 e) + k_2 \dot{r}_1 \\
&= \frac{\partial h}{\partial q_d} \bigg|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial h}{\partial \dot{q}_d} \bigg|_{(q_d, \dot{q}_d)} h(q_d, \dot{q}_d) - \frac{\partial f}{\partial q} \bigg|_{(q, \dot{q})} \dot{q} - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} f(q, \dot{q}) - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} u \\
&\quad - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} d(t) - \dot{u} - \dot{d}(t) + (k_1 + k_2) \dot{r}_1 - k_1^2 r_1 + k_1^3 e \\
&= \underbrace{\frac{\partial h}{\partial q_d} \bigg|_{(q_d, \dot{q}_d)} \dot{q}_d + \frac{\partial h}{\partial \dot{q}_d} \bigg|_{(q_d, \dot{q}_d)} h(q_d, \dot{q}_d) - \frac{\partial f}{\partial q} \bigg|_{(q, \dot{q})} \dot{q} - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} f(q, \dot{q}) - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} u}_{F(\kappa)} \\
&\quad - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} d(t) - \dot{u} - \dot{d}(t) + (k_1 + k_2) (r_2 - k_2 r_1) - k_1^2 r_1 + k_1^3 e \\
&= F(\kappa) - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} d(t) - \dot{u} - \dot{d}(t) + (k_1 + k_2) r_2 - (k_1 + k_2) k_2 r_1 - k_1^2 r_1 + k_1^3 e \\
&= F(\kappa) - \frac{\partial f}{\partial \dot{q}} \bigg|_{(q, \dot{q})} d(t) - \dot{u} - \dot{d}(t) + (k_1 + k_2) r_2 + (-k_1 k_2 - k_2^2 - k_1^2) r_1 + k_1^3 e
\end{aligned}$$

3. NN Approximation (where F and ε are $\mathcal{C}^1$ since h and f are $\mathcal{C}^2$). NOTE: u_d will be defined later!

$$\begin{aligned}
\varepsilon(\kappa) &\triangleq F(\kappa) - \Phi(\kappa, \theta^*) \\
\kappa &\triangleq \begin{bmatrix} q^\top & \dot{q}^\top & q_d^\top & \dot{q}_d^\top & u^\top \end{bmatrix}^\top \in \mathbb{R}^{5n} \\
\kappa_d &\triangleq \begin{bmatrix} q_d^\top & \dot{q}_d^\top & q^\top & \dot{q}^\top & u_d^\top \end{bmatrix}^\top \in \mathbb{R}^{5n}
\end{aligned}$$

4. Controller

$$\dot{u} = (k_1 + k_2 + k_3) r_2 + (1 - k_1^2 - k_2^2 - k_1 k_2) r_1 + k_1^3 e + K_{\text{RISE}} \text{sgn}(r_1) + \Phi(\kappa, \hat{\theta})$$

5. Third:

$$\begin{aligned}
\dot{r}_2 &= \Phi(\kappa, \theta^*) + \varepsilon(\kappa) - \left((k_1 + k_2 + k_3) r_2 + (1 - k_1^2 - k_2^2 - k_1 k_2) r_1 + k_1^3 e + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \right) \\
&\quad - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) + (k_1 + k_2) r_2 + (-k_1 k_2 - k_2^2 - k_1^2) r_1 + k_1^3 e \\
&= \Phi(\kappa, \theta^*) + \varepsilon(\kappa) - \left((\cancel{k_1 + k_2} + k_3) r_2 + \left(1 - \cancel{k_1^2 - k_2^2 - k_1 k_2} \right) r_1 + \cancel{k_1^3 e} + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \right) \\
&\quad - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) + \cancel{(k_1 + k_2) r_2} + \cancel{(-k_1 k_2 - k_2^2 - k_1^2) r_1} + \cancel{k_1^3 e} \\
&= \Phi(\kappa, \theta^*) + \varepsilon(\kappa) - \left(k_3 r_2 + r_1 + K_{\text{RISEsgn}}(r_1) + \Phi(\kappa, \hat{\theta}) \right) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) \\
&= \Phi(\kappa, \theta^*) + \varepsilon(\kappa) - k_3 r_2 - r_1 - K_{\text{RISEsgn}}(r_1) - \Phi(\kappa, \hat{\theta}) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) \\
&= \Phi(\kappa, \theta^*) - \Phi(\kappa, \hat{\theta}) + \varepsilon(\kappa) - k_3 r_2 - r_1 - K_{\text{RISEsgn}}(r_1) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})} d(t) - \dot{d}(t) \\
&\quad - \varepsilon(\kappa_d) + \varepsilon(\kappa_d) + N_0(\kappa_d, \theta^*, \hat{\theta}) - N_0(\kappa_d, \theta^*, \hat{\theta}) + \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) \\
&= \Phi(\kappa, \theta^*) - \Phi(\kappa, \hat{\theta}) + \underbrace{\varepsilon(\kappa) - \varepsilon(\kappa_d)}_{\tilde{N}_\varepsilon(\kappa, \kappa_d)} - k_3 r_2 - r_1 - K_{\text{RISEsgn}}(r_1) \\
&\quad + \left(\underbrace{\frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} - \frac{\partial f}{\partial \dot{q}} \Big|_{(q, \dot{q})}}_{\tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d)} d(t) - \dot{d}(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) + \varepsilon(\kappa_d) \right) \\
&= \Phi(\kappa, \theta^*) - \Phi(\kappa, \hat{\theta}) + \underbrace{\tilde{N}_\varepsilon(\kappa, \kappa_d) + \tilde{N}_2(q, \dot{q}, q_d, \dot{q}_d) d(t)}_{\tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t)} - k_3 r_2 - r_1 \\
&\quad - \underbrace{K_{\text{RISEsgn}}(r_1) - \dot{d}(t) - \frac{\partial f}{\partial \dot{q}} \Big|_{(q_d, \dot{q}_d)} d(t) + \varepsilon(\kappa_d)}_{D(\kappa_d, \theta^*, \hat{\theta}, t)} \\
&= \Phi(\kappa, \theta^*) - \Phi(\kappa, \hat{\theta}) + \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - k_3 r_2 - r_1 - K_{\text{RISEsgn}}(r_1) + D(\kappa_d, \theta^*, \hat{\theta}, t)
\end{aligned}$$

6. Taylor series:

$$\Phi(\kappa, \theta^*) = \Phi(\kappa, \hat{\theta}) + \underbrace{\frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa, \hat{\theta})}}_{\frac{\partial \Phi}{\partial \theta}} \left(\underbrace{\theta^* - \hat{\theta}}_{\tilde{\theta}} \right) + \mathcal{R}(\kappa, \tilde{\theta})$$

7. Fourth:

$$\dot{r}_2 = \frac{\partial \hat{\Phi}}{\partial \tilde{\theta}} \tilde{\theta} + \mathcal{R}(\kappa, \tilde{\theta}) + \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - k_3 r_2 - r_1 - K_{\text{RISEsgn}}(r_1) + D(\kappa_d, \theta^*, \hat{\theta}, t)$$

8. Let $K[\cdot]$ denote the Krasovskii regularization:

$$\dot{r}_2 \in \frac{\partial \hat{\Phi}}{\partial \tilde{\theta}} \tilde{\theta} + \mathcal{R}(\kappa, \tilde{\theta}) + \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) - k_3 r_2 - r_1 - K_{\text{RISE}} K[\text{sgn}](r_1) + D(\kappa_d, \theta^*, \hat{\theta}, t)$$

14 Stability

Let

$$V = \frac{1}{2}e^\top e + \frac{1}{2}r_1^\top r_1 + \frac{1}{2}r_2^\top r_2 + \frac{1}{2}\tilde{\theta}^\top \Gamma^{-1}\tilde{\theta} + P$$

Let

$$\dot{P} \in -\lambda_P P(t) + r_2^\top(t) \left(K_{\text{RISE}} K[\text{sgn}(r_1)] - D(\kappa_d, \theta^*, \hat{\theta}, t) \right)$$

Let

$$\dot{\hat{\theta}} = \text{proj} \left(\Gamma \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa, \hat{\theta})}^\top r_2 \right)$$

Let

$$\tilde{\theta} \triangleq \theta^* - \hat{\theta}$$

Thus,

$$\begin{aligned} \dot{\tilde{\theta}} &= -\dot{\hat{\theta}} \\ &= -\text{proj} \left(\Gamma \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa, \hat{\theta})}^\top r_2 \right) \end{aligned}$$

From Krstic's Appendix E.4,

$$-\tilde{\theta}^\top \Gamma^{-1} \text{proj}(\tau) \leq -\tilde{\theta}^\top \Gamma^{-1} \tau$$

so

$$\begin{aligned} \tilde{\theta}^\top \Gamma^{-1} \dot{\tilde{\theta}} &= \tilde{\theta}^\top \Gamma^{-1} \left(-\text{proj} \left(\Gamma \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa, \hat{\theta})}^\top r_2 \right) \right) \\ &= -\tilde{\theta}^\top \Gamma^{-1} \text{proj} \left(\Gamma \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa, \hat{\theta})}^\top r_2 \right) \\ &\leq -\tilde{\theta}^\top \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa, \hat{\theta})}^\top r_2 \end{aligned}$$

The trajectory derivative

$$\begin{aligned} \dot{V}(z, t) &= e^\top \dot{e} + r_1^\top \dot{r}_1 + r_2^\top \dot{r}_2 + \tilde{\theta}^\top \Gamma^{-1} \dot{\tilde{\theta}} + \dot{P} \\ &= e^\top r_1 - k_1 e^\top e + \cancel{r_1^\top r_2} - k_2 r_1^\top r_1 + r_2^\top \frac{\partial \hat{\Phi}}{\partial \theta} \tilde{\theta} + r_2^\top \mathcal{R}(\kappa, \tilde{\theta}) + r_2^\top \tilde{N}(\kappa, \kappa_d, \theta^*, \hat{\theta}, t) \\ &\quad - k_3 r_2^\top r_2 - \cancel{r_2^\top r_1} - \cancel{K_{\text{RISE}} r_2^\top K[\text{sgn}](r_1)} + r_2^\top D(\kappa_d, \theta^*, \hat{\theta}, t) \\ &\quad - \cancel{\tilde{\theta}^\top \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa, \hat{\theta})}^\top r_2} - \lambda_P P(t) + \cancel{K_{\text{RISE}} r_2^\top K[\text{sgn}](r_1)} - \cancel{r_2^\top D(\kappa_d, \theta^*, \hat{\theta}, t)} \end{aligned}$$

I don't need to finish. This will be stable. However, the update law is clearly not implementable because r_2 is not measurable (since it contains generalized acceleration $\ddot{x}$). That's fine. So we have to do the best we can:

$$\dot{\hat{\theta}} = \text{proj} \left(\Gamma \frac{\partial \Phi}{\partial \theta} \Big|_{(\kappa, \hat{\theta})}^\top r_1 - \sigma_{\text{mod}} \hat{\theta} \right)$$

This update law does not follow from the stability analysis above and that's fine. It's not meant to. We will show it works heuristically.

15 Experiment

Baseline controller is

$$u(t) = \Phi(x(t_{k+1}), \hat{\theta}(t)) + K_P e(t) + K_D \dot{e}(t) + \int_{t_0}^t (K_I e(\tau) + K_{\text{RISE}} \text{sgn}(r_1(\tau))) d\tau$$

Developed controller is

$$u(t) = K_P e(t) + K_D \dot{e}(t) + \int_{t_0}^t \left(K_I e(\tau) + K_{\text{RISE}} \text{sgn}(r_1(\tau)) + \Phi(\kappa(\tau), \hat{\theta}(\tau)) \right) d\tau$$

where

$$\begin{aligned} K_P &\triangleq k_1 k_2 + k_1 k_3 + k_2 k_3 + 1 \\ K_I &\triangleq k_1 k_2 k_3 + k_1 \\ K_D &\triangleq k_1 + k_2 + k_3 \\ \kappa &\triangleq \begin{bmatrix} q^\top & \dot{q}^\top & q_d^\top & \dot{q}_d^\top & u^\top \end{bmatrix}^\top \in \mathbb{R}^{5n} \\ x &\triangleq \begin{bmatrix} q^\top & \dot{q}^\top & q_d^\top & \dot{q}_d^\top \end{bmatrix}^\top \in \mathbb{R}^{4n} \\ e &\triangleq q_d - q \\ r_1 &\triangleq \dot{e} + k_1 e \\ n &= 3 \end{aligned}$$

NOTE: the input to the ResNet in each controller is different! Both use the update law

$$\begin{aligned} \hat{\theta} &= \text{proj}_{\mathcal{U}} \left(\Gamma \left. \frac{\partial \Phi}{\partial \theta} \right|^\top r_1 - \sigma_{\text{mod}} \hat{\theta} \right) \\ \mathcal{U} &\triangleq \overline{B_{\hat{\theta}}}(0_p) \end{aligned}$$

The desired trajectory will be generated by the time invariant (CRITICAL to be time invariant!)

$$\ddot{q}_d = h(q_d, \dot{q}_d)$$

These experiments will be acceleration-level controllers for a quadcopter. Linear accelerations only. We can measure: q is my position, q_d is the target's position. $\dot{q}$ is my velocity, $\dot{q}_d$ is the target's velocity. We cannot measure $\ddot{q}$ or $\ddot{q}_d$.